Email: patents@cumail.in,

This disclosure is submitted in confidence for preliminary assessment of patentability and guidance on further technical development. The project is at the concept/design stage. No implemented prototype, completed dataset, experimental validation, or verified patent search is claimed. The detailed embodiment in Section 6 is a proposed specification, subject to confirmation by the inventors and review by the University IP/patent team. It does not by itself establish novelty, inventive step, medical validity, or patent eligibility.

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Inventor Contribution Statements :

Supervision, coding, data collection, or approval alone does not make a person an inventor. Final inventorship will be decided by the University IP/patent team. Each inventor should edit the statement below so it describes the technical ideas they personally conceived.

Inventor	**Draft contribution statement**
Dayal Chandra Sati	Academic guidance of the project. State here any specific technical idea you personally conceived (for example the constraint-priority or conflict-reporting approach). If none, state "guidance only".
Kuruma Voojola Lokesh	Conception of the nutrition-target and fitness-goal calculation (BMR/TDEE estimation, goal adjustment, macronutrient targets, input validation.
P Vaishnavi	Conception of the Indian-food and dietary-preference data representation (food-record schema, nutrition-data fields, dietary and allergen filtering).
V Surya Teja	Conception of budget-aware selection, serving-cost estimation, and price-aware substitution with recalculation.

A) Please enlist the Key concepts and terms/Keywords/Synonyms used in the Invention?

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S. No.	Keywords for invention which has to be searched	Synonyms / related search terms
1	Budget-aware personalized diet planning	Affordable meal planning; cost-constrained diet recommendation
2	Personalized nutrition targets	Calorie requirements; macronutrient goals; nutrition profiling
3	Indian food recommendation	Common Indian foods; culturally relevant food choices
4	Food-cost estimation	Estimated serving price; meal cost; daily food expenditure
5	Rule-based food selection	Constraint filtering; eligibility rules; food ranking
6	Budget-aware food substitution	Lower-cost alternatives; price-aware replacement
7	Complete daily meal plan	Breakfast, lunch, dinner, snacks; meal-wise nutrition
8	Constraint-priority and conflict reporting	Hard and soft constraints; nutritional deviation; bounded search; shortfall warning
9	Price-data reliability	Price freshness; missing-price handling; dated, location-specific prices

Written Description – Invention Information

1. TITLE OF INVENTION – 15 words max

AI-Enabled Budget-Aware Personalized Diet Planning and Nutritional Recommendation System

In this disclosure, "AI-enabled" means a knowledge-based, rule-driven decision-support workflow for filtering, ranking and substituting foods. No machine-learning model, training process or learned prediction is claimed. The title does not itself establish any technical contribution or patent eligibility.

2. BACKGROUND OF THE INVENTION

Personalized diet recommendation research has explored nutritional requirements, dietary preferences, food restrictions, health-related constraints and food knowledge. The project's Phase 1 literature review discusses Yum-Me, FoodKG, constrained food recommendation over knowledge graphs, health-guided recipe recommendation, ChatDiet and TanSa. These works are useful background, but their goals and personalization factors differ. TanSa is described in the Phase 1 report as an affordable-nutrition recommender that uses price categories and Iranian cuisine. The exact capabilities of each system must be checked against the original papers before any statement about what they lack is relied on.

Cost-constrained diet planning is an established field. Classical work such as the Stigler diet problem, and later linear and integer programming approaches, already combine nutrient requirements with food costs. Combining nutrition with prices or a budget is therefore not claimed as new in itself.

The practical problem addressed here is that a meal plan may be nutritionally appropriate yet financially impractical for a student or other budget-conscious user. Conversely, choosing only low-cost foods without nutrition checks can leave calorie or macronutrient targets unmet. A practical daily plan should consider the user's profile, activity level, fitness goal, dietary restrictions, familiar Indian foods and personal budget together, and should show how each decision was made.

The proposed system uses one transparent workflow. It estimates targets, excludes unsuitable foods, builds a full day of meals, estimates cost, and performs bounded, rule-based lower-cost substitutions. It reports separately whether a generated plan misses targets, whether the search stopped at its limits, and whether data was missing. A heuristic search that fails to find a compliant plan is not presented as proof that none exists.

Food prices are estimates and vary by location, seller, brand and date. A formal patent search has not been conducted or verified. The comparison here rests on the Phase 1 academic review only.

3. BRIEF SUMMARY OF INVENTION

The invention is a proposed software system that generates personalized daily meal plans from the user's age, sex, height, weight, activity level, general fitness goal, dietary preference, exclusions and a daily or monthly food budget. It estimates BMI (descriptive only), BMR and TDEE, then derives goal-based energy and macronutrient targets and acceptance ranges.

A structured food dataset holds serving definitions, nutrient values, dietary class, allergen information, meal-slot roles, and dated, sourced serving-cost estimates. A rule-based engine first removes foods that violate mandatory dietary, exclusion, data-quality or price-admission rules. It then enumerates candidate daily plans within a stated limit, covering breakfast, lunch, dinner and snacks, and ranks them by nutritional-range violation and closeness to targets.

If the selected plan exceeds the budget, a substitution stage replaces components with compatible lower-cost alternatives. Each substitution is evaluated against the whole-day totals. It must strictly reduce cost, must not worsen nutritional-range violation, and must

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not repeat an earlier plan state. The output reports portions, nutrition, estimated cost, budget status, deviations, price-data limitations and search status.

The project is at the concept/design stage. The numeric settings below are proposed values that need inventor confirmation and nutrition-professional review. No dataset, prototype, final algorithm or experimental result is claimed.

4. OBJECTIVE OF THE INVENTION

The objectives of the proposed invention are:

1. Build a validated user profile from age, sex, height, weight, activity level, fitness goal, dietary preference, exclusions, allergy information where supplied, and food budget.
2. Estimate BMI, BMR, TDEE and goal-based calorie and macronutrient targets using documented, configurable equations. BMI is descriptive only.
3. Treat the daily or monthly food budget as an active planning constraint during meal construction and repair, not only as a displayed value.
4. Support commonly consumed Indian foods with vegetarian, egg-containing and non-vegetarian options, subject to user choices.
5. Apply mandatory suitability rules before any food is considered, and never treat missing allergen data as proof of safety.
6. Generate a complete day (breakfast, lunch, dinner, snacks) with meal-wise and daily nutrition and estimated cost.
7. Attempt compatible lower-cost substitutions through defined eligibility, ranking and stopping rules.
8. Report budget excess, nutritional deviations, missing or outdated prices, and search-limit termination separately and accurately.
9. Provide a documented basis for later testing of arithmetic accuracy, rule compliance, budget compliance and recommendation quality.

The system is a general nutrition and fitness-planning aid. It is not medical diagnosis or treatment and does not replace a qualified professional.

5. EXPLAIN PROBLEMS IN PRIOR ART OR EARLIER INVENTION WHICH YOUR PRESENT INVENTION TARGETS TO SOLVE POINTWISE

These are practical problems the proposal targets. They are not assertions that every existing system has these deficiencies.

1. Generic meal plans may not reflect an individual's estimated energy needs, activity level or fitness goal.
2. A nutritionally suitable plan may exceed the user's daily or monthly food budget.
3. Cost-only selection may leave calorie or macronutrient targets substantially unmet.
4. Dietary class, exclusions and allergies reduce the set of usable foods.
5. Recommending single foods or recipes does not necessarily give a complete day with portions, totals and cost.
6. Indian and common foods and local prices may not be central to systems designed for other cuisines or price categories.
7. Prices can be missing, outdated, location-dependent or inconsistent with the serving definition, and must be labelled as estimates.
8. Repeated substitutions can cycle, degrade nutrition, or stop without explanation.
9. A limited heuristic search that fails can be wrongly described as proof that the constraints are impossible.
10. Outputs may hide which rules were enforced and which targets remain unmet.

5A. PROPOSED DISTINGUISHING FEATURES FOR PRIOR-ART ASSESSMENT

(This replaces the earlier "Novel Aspects" heading. The number 5A keeps the form's numbering of Sections 6–12 unchanged. Renumber if your template requires it.)

These are proposed technical features for the IP team to assess. They are not statements of legal novelty or inventive step, and the individual features or their combination may already be known.

1. **Constraint-priority processing.** Dietary class, exclusions, allergen-data completeness and price-admission rules are applied before ranking or substitution and are never relaxed to save money. Nutrition deviation is then minimised, and variety and preference come last.
2. **Whole-plan nutritional-loss control during cost repair.** A substitution is judged on the complete day's totals and is rejected if it increases the defined range-violation measure.
3. **Deterministic substitution ranking.** Eligible swaps are ordered by a fixed list of criteria, with stable tie-breaking.
4. **Cycle-resistant budget repair.** Each accepted swap must strictly reduce cost, and earlier plan states are never revisited.
5. **Price-data admission rules.** Missing or outdated prices are never silently treated as current or as zero.
6. **Separate status reporting.** Budget, nutrition, data and search status are reported independently, and infeasibility is never claimed without proof.

Indian-food coverage is mainly data content and is not asserted as an independent technical feature.

6. DETAILED DESCRIPTION OF THE INVENTION

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6.1 Overall proposed system architecture. The system consists of a user-profile interface, an input-validation module, a nutrition-target module, a structured food and price dataset, an eligibility filter, a bounded plan-selection engine, a cost and substitution engine, and a results module. A web application with a possible Python (Flask or Django) backend and a relational database such as MySQL is contemplated. No particular technology is essential to the workflow, and final choices are open.

6.2 Intended users, inputs and validation. The initial scope is general nutrition planning for healthy adults aged 18–64. Children, pregnant or breastfeeding users, users with diagnosed diseases or therapeutic diets, and users with eating-disorder concerns are out of scope.

Inputs are age, sex category (used only to select the BMR coefficient), height in cm, weight in kg, activity category, goal (maintenance, weight loss or muscle gain), dietary class (vegetarian, egg-containing or non-vegetarian), exclusions, allergy information where supplied, and a daily or monthly budget in INR. Validation requires the following:

- age within 18–64;
- weight within 35–200 kg and height within 120–230 cm;
- activity and goal chosen from the defined lists;
- a positive budget;
- no conflicting dietary declarations.

Invalid input stops generation with a correction message. These limits are configurable.

6.3 Nutrition-target estimation. BMI is calculated for description only, never for diagnosis:

$$BMI = \frac{W}{H_m^2}$$

where

$$W$$

is weight in kg and

$$H_m$$

is height in metres. BMR uses the Mifflin–St Jeor equations, where

$$W$$

is weight in kg,

$$H$$

is height in cm and

$$A$$

is age in years:

$$BMR_{male} = 10W + 6.25H - 5A + 5$$

$$BMR_{female} = 10W + 6.25H - 5A - 161$$

$$TDEE = BMR \times AF$$

The activity factors (

$$AF$$

) are:

Activity level	**Factor**
Sedentary	1.20
Lightly active	1.375
Moderately active	1.55
Very active	1.725
Extra active	1.90

The central energy target is

$$T_E = TDEE,(1+g)$$

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, with proposed goal adjustment

$$g = 0$$

for maintenance,

$$-0.15$$

for weight loss and

$$+0.10$$

for muscle gain. A proposed floor of 1,500 kcal/day for males and 1,200 kcal/day for females is applied with a professional-review notice. A target above 3,500 kcal/day withholds personalised output and refers the user to qualified advice.

The proposed macronutrient split is 20% of energy from protein, 50% from carbohydrate and 30% from fat:

$$T_P = \frac{0.20,T_E}{4}, \qquad T_C = \frac{0.50,T_E}{4}, \qquad T_F = \frac{0.30,T_E}{9}$$

The acceptance ranges are

$$L_E = 0.90,T_E$$

and

$$U_E = 1.10,T_E$$

for energy, and

$$L_j = 0.80,T_j$$

and

$$U_j = 1.20,T_j$$

for

$$j \in {P, C, F}$$

. These are algorithmic acceptance bands, not clinical standards. All parameters (adjustments, floors, split, ranges) are provisional and must be reviewed by a qualified nutrition professional before deployment. Satisfying four tracked quantities does not establish micronutrient adequacy.

6.4 Food and nutrition dataset. Each record is a defined ingredient, standardised recipe portion, packaged item or purchased meal. Raw and cooked quantities are not interchanged without a documented yield conversion.

Field	**Description**
Food ID and version	Stable identifier and revision
Food name	Display name and preparation description
Record type and preparation status	Ingredient / recipe / packaged / purchased meal; raw / cooked / ready-to-eat
Serving quantity and unit	Weight, volume, count or standard portion
Nutrients	Energy, protein, carbohydrate, fat, fibre where available
Dietary class	Vegetarian / egg-containing / non-vegetarian
Ingredient and allergen information	Ingredients, allergen flags, completeness status
Meal slots and role	Permitted meal positions and function (main, base, protein)
Allowed portions	Discrete permitted serving multiples
Estimated serving cost	Cost for the stated serving and cost basis
Price provenance	Date, location, source, currency
Reliability status	Verified / provisional / missing / outdated / excluded
Nutrition provenance and licence	Reference, version, permitted use

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The Indian Food Composition Tables (ICMR–NIN) are a candidate nutrition reference (Phase 2 report). Their licence and permitted reuse must be confirmed before incorporation, and no permission is assumed. The price source, geography and update method are not yet decided.

6.5 Dietary suitability filtering and mandatory rules. A food is eligible only if its dietary class, ingredients, exclusions, price status, nutrient data and serving definition satisfy the user profile and record requirements. If an allergy is declared, a known conflicting ingredient excludes the food, and an incomplete ingredient or allergen record also excludes it from the allergy-filtered set. Even complete records do not guarantee freedom from contamination, so users must verify ingredients themselves.

The following are mandatory and are never overridden to reduce cost: dietary class, explicit exclusions, allergy filtering, permitted portions only, an accepted nutrient and cost basis, and meal-slot and role compatibility.

6.6 Cost and budget model. The initial embodiment uses standardised home-prepared portions. Cost covers the ingredients attributable to the stated portion, including recipe oil and spices where they are part of the documented recipe. Fuel, labour, delivery and equipment are excluded and disclosed as exclusions. A package price is converted to a serving price by:

$$c_{serving} = PackagePrice \times \frac{ServingQuantity}{PackageQuantity}$$

For example, a INR 120 package of 500 g with a 50 g serving gives

$$120 \times 50/500 = 12$$

INR. For a recipe, the serving cost is the sum of ingredient costs divided by the documented number of servings. Daily cost is:

$$C_{day} = \sum_i c_i,x_i$$

where

$$x_i$$

is the number of servings of item

$$i$$

. A monthly budget is converted by:

$$B_{daily} = \frac{B_{monthly}}{D}$$

where

$$D$$

is the planning days confirmed by the user. For example,

$$5400/30 = 180$$

INR per day. This is a daily equivalent only. The system does not claim monthly optimisation, bulk-purchase planning, leftover management or carry-over spending.

Price-admission policy. A price must have a documented source, location, date and compatible serving basis. A price older than 30 days (configurable) is flagged outdated and excluded from budget-compliance calculation until refreshed or replaced by a dated user-confirmed estimate. User-confirmed prices are labelled distinctly, and a missing price is never treated as zero.

6.7 Nutritional deviation measures. For each tracked quantity

$$j \in \{E, P, C, F\}$$

with achieved total

$$N_j$$

, the normalised range violation is:

$$v_j = \frac{\max(L_j - N_j,; 0,; N_j - U_j)}{T_j}$$

and the total is

$$V = \sum_j v_j$$

.

$$V = 0$$

means all four quantities are within range. Closeness to the central targets is:

$$D_N = \frac{1}{4}\sum_{j \in \{E,P,C,F\}} \frac{|N_j - T_j|}{T_j}$$

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Budget excess is

$$E_B = \max(0,; C_{day} - B_{daily})$$

.

6.8 Initial meal selection and portioning. The day has breakfast, lunch, dinner and snack positions. A meal may need several positions (for example a dinner base and a dinner protein). Each position draws only from eligible records with matching slot and role tags, using only permitted portion multiples (such as 0.5, 1.0, 1.5, 2.0). The engine enumerates complete plans in a stable order (position, food ID, portion), up to 10,000 plans. A food ID may appear in at most two positions per day. Meal-wise calorie shares are reported but not enforced in this baseline (an open item).

Candidate plans are ranked lexicographically by:

1. lowest

$$V$$

;
2. lowest

$$D_N$$

;
3. highest declared preference fit;
4. lowest repetition penalty;
5. lowest budget excess;
6. lowest daily cost;
7. stable food-ID and portion order.

The first stage deliberately finds a nutritionally close starting plan, and the repair stage then fits the budget. This baseline is not claimed to find the globally best budget-compliant plan. If the 10,000-plan limit is reached, that fact is recorded in the output.

6.9 Budget-aware substitution. If the initial plan exceeds the budget, the engine considers single-component replacements. A replacement is admissible only if all of the following hold:

- it satisfies all mandatory rules;
- it fits the same meal position and role;
- it reduces plan cost by at least INR 0.01;
- the whole-plan

$$V$$

does not increase;
- the repetition rule is preserved;
- it does not recreate a previously visited plan state.

Totals are recalculated for the whole plan after each trial replacement, so the effect on the full day is evaluated rather than a food-to-food comparison. Admissible swaps are ranked by (1) lowest resulting

$$V$$

, (2) greatest cost saving, (3) lowest resulting

$$D_N$$

, (4) highest preference fit, (5) stable ID and portion order. The best swap is accepted, and the process repeats until the budget is met, no admissible swap remains, or 200 accepted swaps have occurred. Strict cost reduction and stored plan states prevent swapping cycles. The conservative rule may reject a cheaper swap that worsens nutrition, and the system then reports the outcome instead of relaxing the rule.

6.10 Proposed algorithm.

```
INPUT: validated profile, daily budget (or monthly/D), versioned food records, configuration

1. Validate scope, measurements, dietary inputs and budget.
2. Compute BMI, BMR, TDEE, targets T_E, T_P, T_C, T_F and ranges.
3. If target is outside operating domain: withhold plan, show referral guidance, stop.
4. Exclude records failing diet, exclusion, allergen-data, serving,
   nutrient-data or price-admission rules.
5. Build eligible food-and-portion options for each required meal position.
```

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```
6. If any required position has no option: report empty position, stop.
7. Enumerate up to 10,000 complete plans in stable order.
8. Reject plans that violate mandatory or repetition rules.
9. For each valid plan compute nutrient totals, cost, V and D_N.
10. Keep the best plan by the ranking in 6.8; if none valid, report
    "no valid plan generated within the search performed", stop.
11. Store the plan state.
12. While cost > budget and accepted swaps < 200:
      generate admissible single-component replacements (6.9);
      if none: stop repair;
      accept the top-ranked swap; store the new plan state.
13. Report final plan, deviations, budget status, price limitations, search flags.
```

6.11 Output, conflict handling and priority order. The output states four things separately:

- **Budget status:** within or above the estimated daily budget.
- **Nutrition status:** within or outside each tracked range.
- **Data status:** accepted, user-entered or outdated prices, and missing data.
- **Search status:** normal end, no admissible swap, or search limit reached.

Examples of permitted messages: "Generated plan is within the estimated budget and the four tracked nutritional ranges", "Generated plan exceeds the estimated budget by the stated amount", and "No budget-compliant plan was found by this search within its stated limits."

The initial embodiment contains no infeasibility-proof mechanism, so the system never states that no feasible plan exists. If compromises are needed, the priority order is:

1. allergy and safety exclusions;
2. dietary class and explicit restrictions;
3. valid meal and portion rules;
4. the budget;
5. nutrition deviation;
6. preference and variety.

6.12 Worked example (illustrative example using hypothetical data, not an experimental result or dietary advice).

The synthetic records below exist only to show the arithmetic and are not verified food-composition values or market prices. They must not populate a production dataset. Price date and location are "hypothetical demonstration only", reliability is "demonstration, not production eligible", and all are single-portion cooked records.

ID	Illustrative portion	kcal	Protein g	Carb g	Fat g	Diet	INR	Slot / role
H01	Oats-and-milk breakfast	620	28	82	20	Vegetarian	40	Breakfast / main
H02	Rice-and-dal lunch	850	38	116	26	Vegetarian	65	Lunch / main
H03	Roti-and-vegetable base	550	30	76	14	Vegetarian	30	Dinner / base
H04	Paneer-based protein dish	360	24	12	24	Vegetarian	60	Dinner / protein
H05	Soy-based protein dish	180	24	12	4	Vegetarian	30	Dinner / protein
H06	Fruit-and-groundnut snack	360	20	52	8	Vegetarian	15	Snack / main

Fictional user. Male coefficient, age 21, 70 kg, 175 cm, moderately active, maintenance goal, vegetarian, no declared allergies, daily budget INR 180.

Step 1: BMI, BMR and TDEE.

$$BMI = \frac{70}{1.75^2} \approx 22.86$$

$$BMR = 10(70) + 6.25(175) - 5(21) + 5 = 700 + 1093.75 - 105 + 5 = 1693.75$$

$$TDEE = 1693.75 \times 1.55 = 2625.3125$$

Step 2: targets (maintenance, so

$$g = 0$$

and

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$$T_E = 2625.3125$$

kcal):

$$T_P = \frac{0.20 \times 2625.3125}{4} = 131.27 \text{ g}, \quad T_C = \frac{0.50 \times 2625.3125}{4} = 328.16 \text{ g}, \quad T_F = \frac{0.30 \times 2625.3125}{9} = 87.51 \text{ g}$$

Quantity	Central target	Acceptance range
Energy	2,625.31 kcal	2,362.78 – 2,887.84 kcal
Protein	131.27 g	105.01 – 157.52 g
Carbohydrate	328.16 g	262.53 – 393.80 g
Fat	87.51 g	70.01 – 105.01 g

Step 3: initial plan. In this demonstration the candidate set contains H01, H02, H03, H06 and either H04 or H05 for the dinner protein. Both options give

$$V = 0$$

, so ranking falls to

$$D_N$$

. This is approximately 0.0479 with H04 and approximately 0.0747 with H05, so H04 is selected.

Meal	Records	kcal	Protein g	Carb g	Fat g	INR
Breakfast	H01	620	28	82	20	40
Lunch	H02	850	38	116	26	65
Dinner	H03 + H04	910	54	88	38	90
Snack	H06	360	20	52	8	15
Total		**2,740**	**140**	**338**	**92**	**210**

All four totals are inside their ranges (

$$V = 0$$

), but the cost exceeds the budget by INR 30.

Step 4: substitution H04 → H05. Both have the same diet class, dinner position and protein role. The replacement saves INR 30, leaves protein and carbohydrate unchanged, and reduces energy by 180 kcal and fat by 20 g. The new totals stay inside all ranges, so

$$V$$

stays 0 and the swap is admissible.

Result	Initial plan	After substitution
Energy	2,740 kcal	2,560 kcal
Protein	140 g	140 g
Carbohydrate	338 g	338 g
Fat	92 g	72 g
Estimated cost	INR 210	INR 180
Budget excess	INR 30	INR 0
Range violation V	0	0

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Final message: "Generated demonstration plan fits the estimated INR 180 daily budget and the four specified nutritional ranges after one substitution. Values are hypothetical; micronutrient adequacy, allergy safety and clinical suitability have not been established."

If the budget had been INR 150, no admissible swap would remain in this tiny candidate set. The system would then report that the plan exceeds the budget by INR 30 and that no budget-compliant plan was found within the stated search. It would not say that none exists.

6.13 Data, privacy and safety. The proposed interface should explain why data is collected, obtain consent, and collect only what planning needs. It should use access control, encrypted transmission, defined retention, and deletion on request. Diagnostic labels and medical histories should not be collected. Allergy data is handled conservatively. Outputs state that nutrition and cost are estimates, make no diagnosis or treatment claim, and refer users with medical conditions or severe allergies to qualified professionals. These are proposed requirements and not implemented controls.

6.14 Development status, open decisions and limitations. The project is a proposed concept. No prototype, final dataset, price feed, architecture drawing in code, test set or experimental result exists. The following decisions remain **unresolved**:

- nutrition-professional review of targets, floors and ranges;
- price source, geography and update method;
- licence confirmation for IFCT/ICMR-NIN data;
- search limits and tolerance tuning;
- fractional portions and meal-wise calorie shares;
- treatment of fibre and micronutrients;
- technology choices.

Planned testing covers arithmetic checks, dietary-rule enforcement, missing-price behaviour, substitution eligibility, cycle prevention, search-limit reporting and reproducibility. Performance claims require an implemented system and a documented evaluation.

PROPOSED TECHNICAL CLAIM THEMES FOR PATENT-TEAM REVIEW

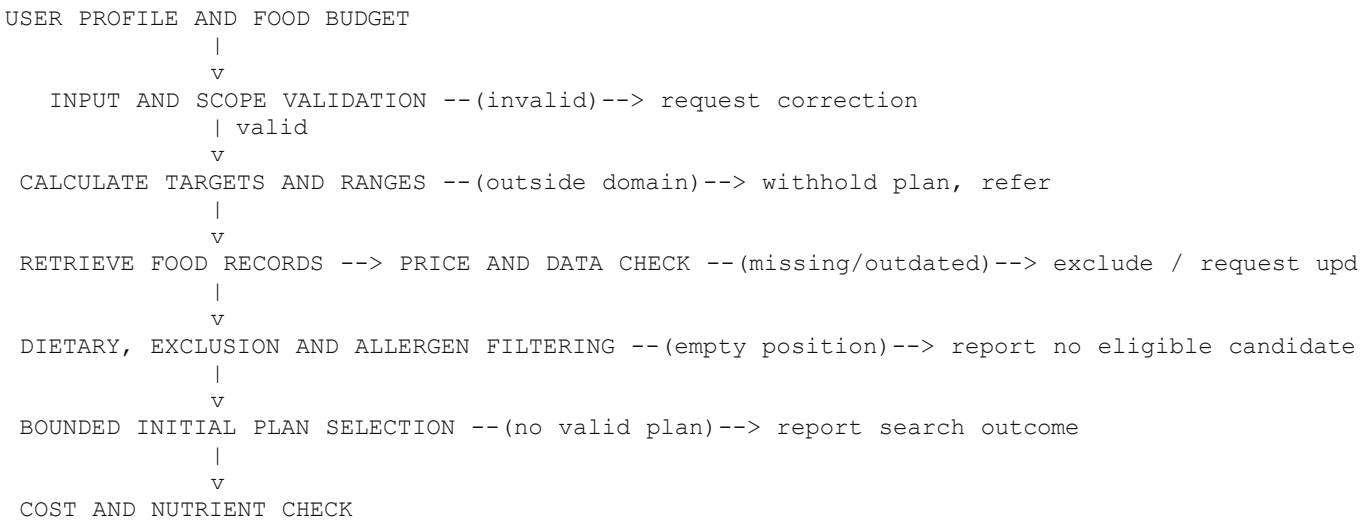
These are discussion themes, not filing-ready legal claims.

1. A computer-implemented workflow that receives a user profile and budget, calculates documented nutrition targets, and forms eligible meal-position candidates using dietary and data-quality rules.
2. A bounded plan-selection process that ranks candidate plans by a range-violation measure and a central-target deviation measure.
3. A budget-repair process that tests lower-cost replacements against whole-plan nutrient totals and rejects any replacement that increases the range-violation measure.
4. Deterministic substitution ranking with strict cost reduction, visited-state exclusion and explicit stopping conditions.
5. Separate reporting of budget, nutrition, data and search status, without asserting unproven infeasibility.
6. A price-admission step based on source, date, location and serving basis.

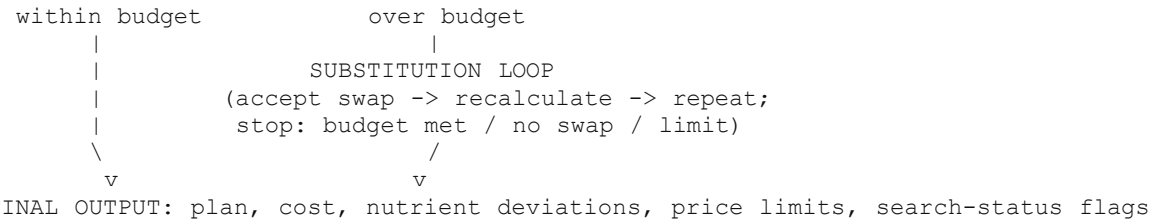
Method, system (processor and memory) and computer-readable-medium categories may be considered if supported. The IP team is specifically asked to assess eligibility under **Section 3(k) of the Indian Patents Act** (computer programme per se, algorithms, mathematical and business methods). The words "AI", "processor" or "database" are not relied on to establish eligibility.

7. DRAWINGS OR SUPPORT MATERIAL

Figure 1 is a proposed workflow. It is a design illustration, not a screenshot or evidence of implementation. The signed contribution statements, dated conception records and project reports can serve as supporting material (see Section 6 and Section 11).



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8. SUMMARY OF THE FIGURES / DIAGRAMS / DRAWINGS

Figure 1 shows profile and budget collection, validation, target calculation, data admission checks, dietary filtering, bounded initial selection, cost and nutrition evaluation, budget-aware substitution and final reporting. The branches cover invalid input, targets outside the operating domain, missing or outdated prices, empty candidate positions, unsuccessful plan generation and substitution termination. It is a functional view of the proposed workflow and does not represent completed implementation or validation.

9. ABSTRACT (150 words max; 138 words)

This disclosure describes a proposed software system for personalized daily meal planning under dietary, nutritional and estimated-cost constraints. A user profile supplies body measurements, activity level, fitness goal, dietary choices, exclusions and a daily or monthly food budget. Documented equations derive energy and macronutrient targets. A structured food dataset provides serving definitions, nutrient values, dietary and allergen information, meal roles and dated price estimates. A rule-based engine filters ineligible foods and performs bounded selection of breakfast, lunch, dinner and snack portions. When the plan exceeds the budget, compatible lower-cost substitutions are evaluated against whole-plan nutritional ranges, mandatory restrictions and cycle-prevention rules. The output reports portions, nutrition, estimated cost, budget status, deviations and search-limit warnings; failure to find a plan is not presented as proof of infeasibility. The project is at concept stage; no prototype or validation is claimed.

10. BRIEF DETAIL OF PARTS OF THE DEVICE, APPARATUS, PRODUCT IN BULLET POINTS

The invention is software-based, so these are functional modules rather than physical parts.

- **User-profile module:** collects measurements, activity, goal, dietary choices, exclusions and budget.
- **Validation module:** checks fields, units, scope and operating limits.
- **Nutrition-calculation module:** computes BMI, BMR, TDEE, energy and macronutrient targets and ranges.
- **Food and price dataset:** versioned serving definitions, nutrients, dietary and allergen data, meal roles, cost and provenance.
- **Price-admission module:** checks source, date, location and serving basis before a price is used.
- **Suitability filter:** removes records that break mandatory rules.
- **Selection engine:** enumerates and ranks bounded candidate daily plans.
- **Cost and substitution engine:** costs plans, tests lower-cost swaps on whole-plan totals, and blocks repeated states.
- **Results module:** shows meals, portions, nutrition, cost, deviations and status flags.

11. BRIEFLY EXPLAIN THE STEPS, FLOW CHART, FUNCTIONALITY OF INVENTION RELATED TO SOFTWARE

This describes proposed functionality, not completed implementation. The workflow is shown in Figure 1 and specified in Section 6.

1. Collect the profile, dietary choices, exclusions, goal and budget, and validate them.
2. Convert a monthly budget to a confirmed daily equivalent where needed.
3. Calculate BMI (descriptive), BMR, TDEE, targets and ranges, and withhold output if outside the operating domain.
4. Retrieve records and apply dietary, allergen, data and price-admission filters.
5. Form eligible food-and-portion options per meal position.
6. Enumerate and rank candidate plans within the declared limit.
7. Calculate meal-wise and daily nutrition, cost,

   $$V$$

   and

   $$D_N$$

   .
8. If over budget, evaluate admissible lower-cost swaps on whole-plan totals within the swap limit.
9. Stop on budget fit, no admissible swap, or the search limit.
10. Display the plan with separate budget, nutrition, data and search status, and state clearly that it is an estimate and not medical advice.

12. BRIEFLY EXPLAIN THE STEPS OF PROCESS, COMPOSITION IN BULLET POINTS

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The invention is a software process. No physical composition is claimed.

- Document the intended population, equations, operating limits, ranges, portion rules and search bounds.
- Prepare licensed food and price records with serving definitions, provenance and reliability status.
- Validate inputs and derive transparent targets.
- Apply mandatory filters and generate candidate meal combinations.
- Calculate nutrition and cost, and apply permitted lower-cost swaps where needed.
- Report deviations, budget excess, missing data and search termination accurately.
- Test against defined scenarios before making any performance claim.

COMPARISON: PRIOR ART vs. NEW INVENTION

This is a preliminary framework. Source-specific points must be verified against the original documents.

Aspect	Prior art / existing approaches	Present invention (proposed system)
Cost-constrained diet planning	Established in classical diet optimisation (Stigler) and linear and integer programming	Does not claim nutrients, costs and a budget together as new
Personalised nutrient recommendation	Yum-Me is relevant to nutrient-based meal recommendation	Target estimation, then full-day assembly and budget repair
Food knowledge	FoodKG is relevant to structured food knowledge	Versioned serving records with eligibility metadata, meal roles and dated costs; no knowledge graph claimed
Conversational nutrition	ChatDiet is relevant to LLM-assisted recommendation	Deterministic rule-based baseline; no LLM or trained model
Affordable recommendation	TanSa is described in Phase 1 as using price categories for Iranian cuisine; exact method to be verified	Individual daily limit with explicit replacement rules; distinctiveness unverified
Substitution	Nutrient-aware and cost-aware substitution may already be known	Whole-plan range control, deterministic ranking, strict cost reduction, visited-state exclusion
Price data	Handling varies; needs source-specific review	Source, location, date and admission rules
Conflict reporting	Exact solvers and heuristics support different conclusions	Bounded heuristic result is never presented as proof of infeasibility
Food context	Systems cover different cuisines	Indian/common food focus planned; data content alone is not claimed as technical
Implementation evidence	Published systems report their own evaluations	Concept stage; no prototype, dataset or results
Patent status	No patent search verified	Formal search, novelty, inventive-step and Section 3(k) review requested

RELATED PUBLICATION

Project publications and disclosures: None.

Research literature identified in the project review (academic publications, not patents; they can still be relevant prior art; verify details and add DOIs against the originals):

1. S. Abhari et al., "Designing and evaluating a nutrition recommender system for improving food security in a developing country," *Archives of Iranian Medicine*, vol. 26, no. 11, pp. 629–641, 2023.
2. L. Yang et al., "Yum-Me: A personalized nutrient-based meal recommender system," *ACM Transactions on Information Systems*, vol. 36, no. 1, Art. 7, 2017.
3. S. Haussmann et al., "FoodKG: A semantics-driven knowledge graph for food recommendation," *The Semantic Web – ISWC 2019*, pp. 146–162.
4. Z. Yang et al., "ChatDiet: Empowering personalized nutrition-oriented food recommender chatbots through an LLM-augmented framework," *Smart Health*, vol. 32, Art. 100465, 2024.
5. G. J. Stigler, "The Cost of Subsistence," *Journal of Farm Economics*, vol. 27, no. 2, pp. 303–314, 1945 (classical cost-constrained diet problem).

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Related patents / formal patent search:

No verified patent results are asserted. A formal patent search has not yet been conducted.

Suggested search phrases:

- "budget-constrained diet planning";
- "cost-aware meal planning";
- "food substitution price nutrition";
- "nutrition optimization food budget";
- "personalized meal plan cost".

The University IP/patent team is requested to run the formal search and assess novelty, inventive step, Section 3(k) eligibility, sufficiency of disclosure, inventorship and ownership before any filing decision.

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